

Date ___/___/___

Binary and Decimal Number System

(I) Decimal to Binary:

method-1

rule:

- (i) - n Divided by 2
- (ii) - store remainder
- (iii) - repeat above two steps till $n \neq 0$
- (iv) - reverse the number

example, $n = 10$

answer = 0101

reverse the number

$$(1010)_2 = (10)_{10}$$

Division	Rem
$10/2 = 5$	0
$5/2 = 2$	1
$2/2 = 1$	0
$1/2 = 0$	1

$$\begin{array}{|c|c|c|c|} \hline 3 & 2 & 1 & 0 \\ \hline 2 & 2 & 2 & 2 \\ \hline \end{array}$$

$$(1010)_2 = (10)_{10}$$

method-2

$\left. \begin{array}{l} \text{bit} = n \& 1; \\ n = n >> 1; \end{array} \right\} \text{ till } n \neq 0$

answer = 0;

while ($n \neq 0$) {

bit = $n \& 1$;

$n = n >> 1$;

answer = $(10^0 \times \text{digit}) + \text{answer}$.

for finding reverse

$$\text{ans} = (\text{digit} \times 10^i) + \text{ans}$$

int range $[-2^{31}, 2^{31}-1]$

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उत्तर some plan में sequence में number बनाना है,

$$ans = (ans \times 10) + digit$$

[1, 2, 3]



digit making

same plan

$$\begin{aligned} (0 \times 10^0) + 1 &= 1 \\ (1 \times 10) + 2 &= 12 \\ (12 \times 10) + 3 &= 123 \end{aligned}$$

reverse.

$$\begin{aligned} (1 \times 10^0) + 0 &= 1 \\ (2 \times 10^1) + 1 &= 21 \\ (3 \times 10^2) + 21 &= 321 \end{aligned}$$

$$ans = (ans \times 10) + digit$$

$$ans = (digit \times 10^i) + ans$$

Binary conversion for -ve numbers:

- ignore the -ve part
- find the 2's complement (1's complement + 1)

example $n = -6$

step 1 = $n = 6$

step 2 =

00000 - - 0110
1's 11 .. 11001
+1

11 .. 11010

at the time of printing

2's complement of above result.

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Ques: write a code for conversion of decimal to binary for -ve number

II) Binary to Decimal:

$$\begin{array}{|c|c|c|c|c|} \hline 1 & 0 & 1 & 0 & 1 \\ \hline 2^4 & 2^3 & 2^2 & 2^1 & 2^0 \\ \hline \end{array}$$

$$= 2^4 + 2^2 + 2^0 = 16 + 4 + 1 = 21$$

check.

$$(10101)_2 = (21)_{10}$$

code

```
while (n != 0) {
    int bit = n & 1;
    int digit = n % 10;
    if (bit == 1) {
        ans += 2i;
        int
    }
    n = n / 10;
    i++;
}
```

Division	Rem
21/2 = 10	1
10/2 = 5	0
5/2 = 2	1
2/2 = 1	0
1/2 = 0	1